NUMERICAL METHODS FOR PDES
Lecture 4

Weak derivatives

$f : [0,1] \longrightarrow \mathbb{R}$

$g : [0,1] \longrightarrow \mathbb{R}$ is a weak derivative of f if

$$\int_0^1 g(x)\,\phi(x)\,dx = -\int_0^1 f(x)\,\phi'(x)\,dx < +\infty$$

$\forall\ \phi \in C_0^\infty((0,1))$

↑ compact support

$\{x \in (0,1) : \phi(x) \neq 0\}$

classical $\Rightarrow$ weak

$$H^1(0,1) = \{ v \in L^2(0,1) : v' \in L^2(0,1) \}$$

↑ weak deriv.

$$H_0^1(0,1) = \{ v \in H^1(0,1) : v(0) = 0 = v(1) \}$$

(WP): Find $u \in V := H_0^1(0,1)$

$$\int_0^1 u' v' \, dx = \langle f, v \rangle \qquad \forall v \in V$$

where $f \in V'$ ← dual of V

- $f \in L^2(0,1)$ $\quad \langle f, v \rangle = \int_0^1 f v \, dx$
- $f = \delta_{1/2}$ $\quad \langle f, v \rangle = v(1/2)$

(WP) is well posed !

$(\exists!\ \text{sol} + |u|_1 \le \|f\|_{V'})$

$$\|f\|_{V'} = \sup_{v \in V} \frac{\langle f, v \rangle}{\|v\|_V}$$

Theorem u Solves (WP) $\iff$

u is the unique solution to the minimization problem:

Find $u \in V$: $J(u) \le J(v)\ \forall v \in V$

Where: $J(v) = \frac{1}{2}\int_0^1 |v'|^2 - \langle f, v \rangle$

$=: \frac{1}{2} A(v,v) - F(v)$

where $A(u,v) = \int_0^1 u'v' dx \quad \forall u, v \in V$

$$A(v,v) = \int_0^1 v'v' dx = |v|_1^2$$

Note: In $V = H_0^1(0,1)$ $\|v'\|_0^2$

$|\cdot|_1$ is a norm

(while it is only a seminorm over $H^1(0,1)$; the standard norm of $H^1(0,1)$ is

$$\|v\|_1 = \left(\|v\|_0^2 + |v|_1^2 \right)^{1/2}$$

$\int_0^1 v^2 dx$ $\int_0^1 v'^2 dx$

Proof: "$\Leftarrow$" (see L3)

"$\Rightarrow$" Assume u solves (WP),

$\forall\, w \in V$,

$$J(u+w) = \frac{1}{2} a(u+w, u+w) - F(u+w)$$

$$\| \quad v \in V$$

$$= \frac{1}{2} a(u,u) + a(u,w) + \frac{1}{2} a(w,w) - F(u) - F(w)$$

$a(u,w) - F(w) = 0$ by (WP)

$$= J(u) + \frac{1}{2} a(w,w) \geq J(u)$$

$\| \quad 0$

□

Remark: The above generalises to non-homogeneous boundary

conditions. Indeed, consider the problem

$$\begin{cases} -u'' = f & \text{in } (0,1) \\ u(0)=\alpha \,;\; u(1)=\beta \end{cases}$$

Let $\tilde{u} \in H^1(0,1)$ a "lifting", that is, $\tilde{u}(0)=\alpha$ $\tilde{u}(1)=\beta$.

Then, we consider the affine space

$$\tilde{u} + \underbrace{H_0^1(0,1)}_{\text{space of variations}}$$

In practice, we want

$u \in H^1(0,1)$: $u(0)=\alpha$; $u(1)=\beta$:

$$= a(u,v) = \langle f, v \rangle$$

$$\| = a(\tilde{u}+\hat{u}, v)$$

$$\Rightarrow a(\hat{u}, v) = \underbrace{\langle f, v \rangle - a(\tilde{u}, v)}_{\tilde{F}(v)}$$

①

Find $\hat{u} \in V = H_0^1(0,1)$:

$$a(\hat{u}, v) = \tilde{F}(v) \quad \forall v \in V$$

② Set $u = \tilde{u} + \hat{u}$

Galerkin method: Given (WP) set on the ∞-dim. space V, consider $V_h < V$ of finite dimension, and solve:

Find $u_h \in V_h$: $A(u_h, v_h) = (u_h, v_h)$ $\forall v_h \in V_h$

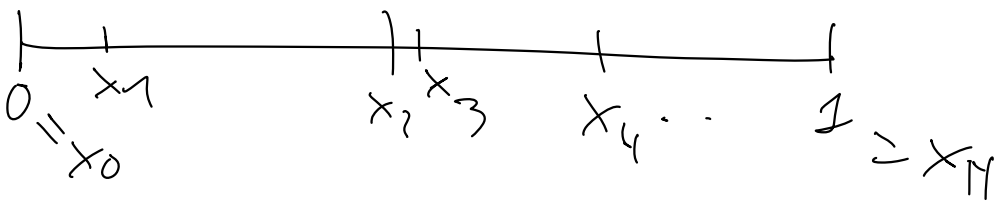
Finite Element Method (FEM)

defines V_h to be made of piecewise polynomial functions

Example: (piecewise) linear FEM (Courant)

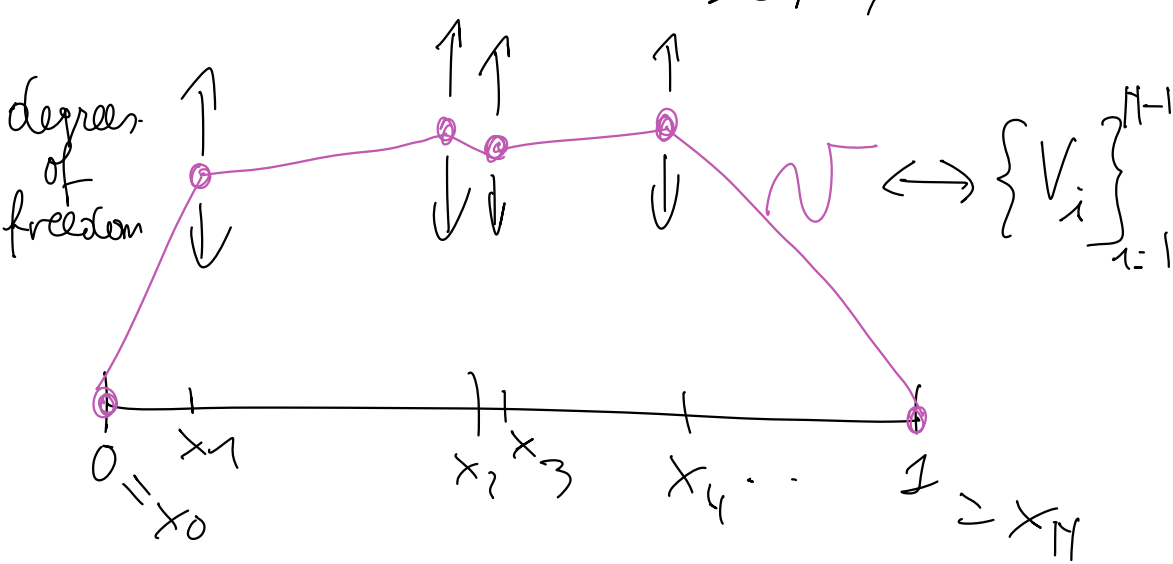
① fix a partition of $\Omega = (0,1)$:

$$0 = x_0 < x_1 < \cdots < x_M = 1$$



② $V_h = \{ v \in H_0^1(0,1) : v|_{[x_{i-1}, x_i]} \in \mathbb{P}_1([x_{i-1}, x_i]$

$i = 1, \ldots, M$



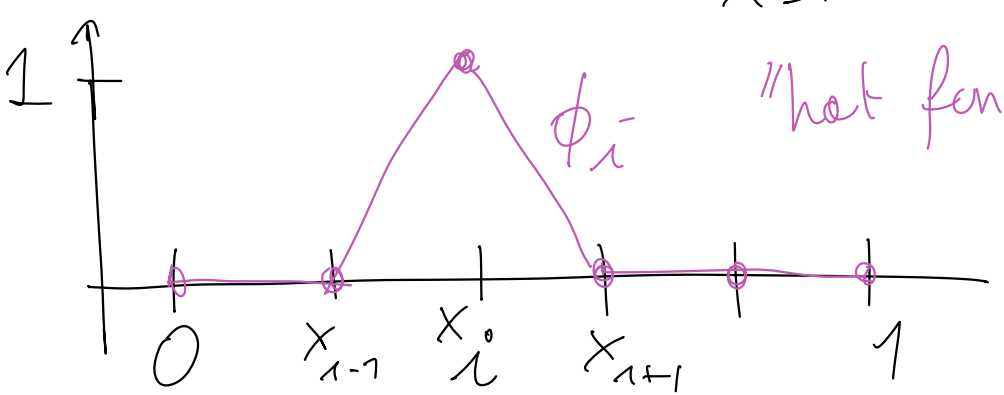
(3) : linear FEM : Find $u_h \in V_h$:

$$a(u_h, v_h) = F(v_h) \quad \forall v_h \in V_h$$

Algebric form of the FEM

Simplicity, assume $\forall i \; x_i - x_{i-1} \equiv h$ (uniform partition)

$$V_h = \text{span}\left\{ \phi_i \right\}_{i=1}^{M-1} \text{ where}$$



$$\phi_i(x) = \begin{cases} \frac{x - x_{i-1}}{h} & x \in (x_{i-1}, x_i] \\ \frac{x_{i+1} - x}{h} & x \in (x_i, x_{i+1}] \\ 0 & \text{otherwise} \end{cases}$$

$$\phi_i'(x) = \begin{cases} 1/h \\ -1/h \\ 0 \end{cases}$$

$$u(x) = \sum_{j=1}^{M-1} U_j \phi_j(x)$$

↖ degrees of freedom (DoF) of linear FEM

FEM linear $\Rightarrow$ sufficient to test with ϕ_i $\forall i = 1, \dots, M-1$

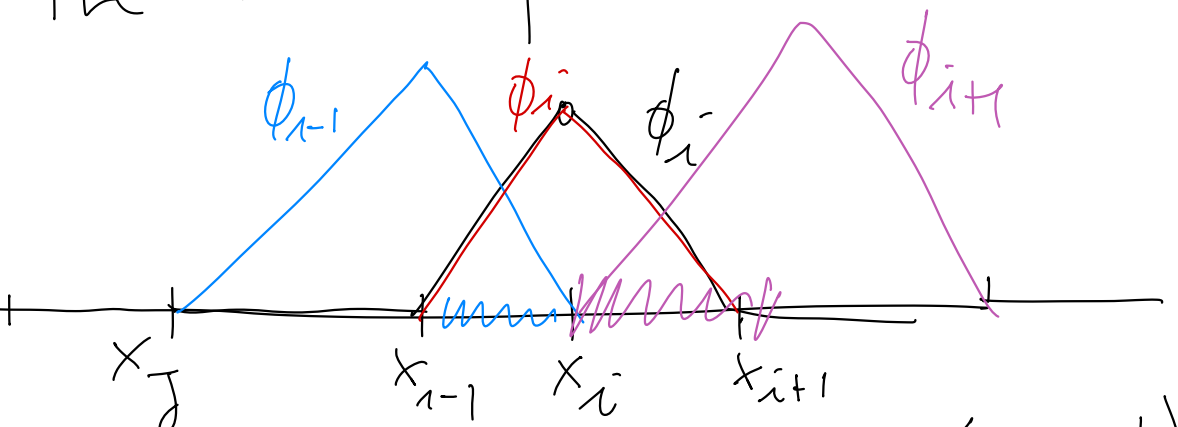
$\Rightarrow$ FEM $\Leftrightarrow$

Find $U = \{U_j\}_{j=1}^{M-1} \in \mathbb{R}^{M-1}$:

$$\sum_{j=1}^{M-1} U_j \int_0^1 \phi_j' \phi_i' = \int_0^1 f \phi_i \, dx$$

$\forall i = 1, \dots, M-1$

The i-th equation :



$$U_{i-1}\left(\int_0^1 \phi_{i-1}' \phi_i'\right) + U_i \left(\int_0^1 \phi_i' \phi_i'\right) + U_{i+1}\left(\int \phi_{i+1}' \phi_i'\right)$$

$$U_{i-1}\int_{x_{i-1}}^{x_i} \phi_{i-1}' \phi_i' + U_i\left(\int_{x_{i-1}}^{x_i} \phi_i' \phi_i' + \int_{x_i}^{x_{i+1}} \phi_i' \phi_i'\right) + U_{i+1}\left(\int_{x_i}^{x_{i+1}} \phi_{i+1}' \phi_i'\right) = \int_{x_{i-1}}^{x_i} f\phi_i + \int_{x_i}^{x_{i+1}} f\phi_i$$

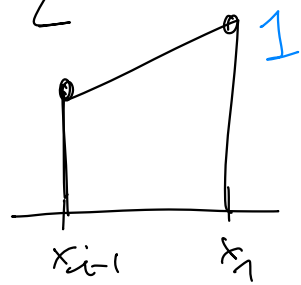
$-1/h$ $1/h$ $\quad$ $1/h$ $1/h$ $\quad$ $-1/h$ $-1/h$ $\quad$ $1/h$ $-1/h$

$$-\frac{1}{h}U_{i-1} + \frac{2}{h}U_i - \frac{1}{h}U_{i+1} = \int_{x_{i-1}}^{x_i} f\phi_i - \int_{x_i}^{x_{i+1}} f\phi_i$$

Let us consider trapezoidal quadrature rule:

$$\int_{x_{i-1}}^{x_i} f\,\phi_i \approx h\left[\underbrace{f(x_{i-1})\,\phi_i(x_{i-1}) + f(x_i)\,\phi_i(x_i)}_{2}\right]$$

$\phi_i(x_{i-1}) = 0$, $\phi_i(x_i) = 1$



$$= \frac{h}{2} f(x_i)$$

$$\int_{x_1}^{x_{i+1}} f\,\phi_i \approx \frac{h}{2} f(x_i)$$

$$\Rightarrow$$

$$-\frac{1}{h} U_{i-1} + \frac{2}{h} U_i - \frac{1}{h} U_{i+1} = h\, f(x_i)$$

FEM + trapez. rule

III

$$\frac{-v_{i-1} + 2v_i - v_{i+1}}{h^2} = f(x_i)$$

$$\Updownarrow$$

FD method !!

So: In one dimension, for the problem $-u'' = f$, if the partition is uniform

FD $\equiv$ FEM + trapezoidal rule